Econ 814 Recitation 3 Notes

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1 Coding Tauchin's method

Let's start with understanding the goal behind Tauchin's method. In these types of models, agents face a continuous stochastic income process. A typical example would be assuming that income is given by an AR1:

$$z_{t+1} = \alpha + \rho z_t + \sigma \xi_{t+1}$$

Where $\epsilon_{t+1} \sim \mathbb{N}(0, 1)$

Since this AR1 is continuous, it can take on infinite number of values. In order to compute the value and policy functions, we would need to discretize it. We can do so via a Markov chain, and this process is known as Tauchin's method.

1.1 Choose an evenly spaced grid

- $X = \{e_1, ..., e_n\}$ will be the state space of the Markov chain
- The elements on the ends will be given by

$$e_1 = \mu_z - m\sigma_z$$

$$e_n = \mu_z + m\sigma_z$$

  where $\mu_z = \frac{\alpha}{1-\rho}$ and $\sigma_z = \frac{\sigma}{(1-\rho^2)^{1/2}}$

- The other points are evenly spaced between these two

$$e_i = e_1 + \Delta(i-1),\ \Delta = \frac{2m\sigma_z}{n-1}$$

1.2 Compute the transition matrix

- Let $\epsilon = e_j - e_{j-1}$
- Assign individual elements of P as follows

$$p_{ij} = \begin{cases} \Phi(\frac{e_1 + \epsilon/2 - \alpha - \rho e_i}{\sigma}) & j = 1 \\ \Phi(\frac{e_j + \epsilon/2 - \alpha - \rho e_i}{\sigma}) - \Phi(\frac{e_j - \epsilon/2 - \alpha - \rho e_i}{\sigma}) & 1 < j < n \\ 1 - \Phi(\frac{e_1 - \epsilon/2 - \alpha - \rho e_i}{\sigma}) & j = n \end{cases}$$

1.3 Defining a function for Tauchin's method

First, we need define the state space. Let's choose 5 states based on the following model

$$\log(w_{t+1}) = 0.7\log(w) + \epsilon$$

Where $\epsilon \sim \mathbb{N}(0, 0.1)$

We can define our variables here to be $\rho = 0.7$, $\sigma = 0.1$, $\alpha = 0$

To get $e_1, ..., e_n$ we need to calculate μ_z and σ_z. We can do that by

```
import numpy as np
#define parameters
rho = 0.7
sigma = 0.1
a = 0
mu_z = a / (1-rho)
sigma_z = sigma / np.sqrt(1-rho**2)
```

Then, we calculate e_1 through e_n

```
X = np.zeros(5) #create an array of 0's, 5 spaces long
X[0] = mu_z - 2*sigma_z #create the first state
X[n-1] = mu_z + 2*sigma_z #create the last state
Delta = (X[n-1]-X[0])/(n-1) #create the distances in between
for i in range(1,n-1):
X[i] = X[i-1]+Delta
```

Now we have an array, X, with our state space. Next is to calculate the probability matrix P

```
from scipy.stats import norm
P = np.zeros((n,n)) #create a matrix of 0's for the transition
    matrix
for i in range(n): #loop over the rows
    ez = a + rho*X[i]
    P[i,0] = norm.cdf((X[0]+Delta/2-ez)/sigma) #probability for
        first column
    for j in range(1,n-1): #loop over the columns
        P[i,j] = norm.cdf((X[j] + Delta/2 - ez) / sigma) -
            norm.cdf((X[j] - Delta/2 - ez) / sigma)
            #probability for middle columns
    P[i, n-1] = 1 - norm.cdf((X[n-1] - Delta/2 - ez) / sigma)
        #probability for last column
```

1.4 Setting Parameters and testing

Now that we have the P matrix set, and the state space defined, we can wrap it all up into a function:

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt


def S_tauchen(n, m, a, rho, sigma):
    X = np.zeros(n)
    mu_z = a / (1-rho)
    sigma_z = sigma / np.sqrt(1-rho**2)
    X[0] = mu_z - m * sigz
    X[n-1] = mu_z + m * sigz
    Delta = (X[n-1] - X[0]) / (n - 1)
    for i in range(1, n - 1):
        X[i] = X[i-1] + Delta
    P = np.zeros((n, n))
    for i in range(n):
        ez = a + rho * z[i]
        P[i, 0] = norm.cdf((X[0] + Delta/2 - ez) / sigma)
        for j in range(1, n - 1):
            P[i, j] = norm.cdf((X[j] + Delta/2 - ez) / sigma) -
                norm.cdf((X[j] - Delta/2 - ez) / sigma)
        P[i, n-1] = 1 - norm.cdf((z[n-1] - Delta/2 - ez) /
            sigma)
    return X, P


#define parameters
rho = 0.7 #income shock persistence
sigma = 0.1 #income shock stdev
a = 0
ny = 5 # number of gridpoints over income shocks

X, P = S_tauchen(ny, (ny-1)/2, a, rho, sigma)
```

2 Example with two sector model

[1]Consider a two-sector version of the neoclassical growth model, where one sector produces the consumption good c_t with technology

$$c_t = z_t^c f\left(k_t^c, n_t^c\right)$$

[1]Note thanks to Professor Gianluca Violante for this example!

and the other sector produces the investment good i_t with technology

$$i_t = z_t^i f\left(k_t^i, n_t^i\right).$$

Assume f is CRS, strictly concave and increasing in both argument. Note that f is the same in both sectors.

Aggregate capital is $k_t = k_t^c + k_t^i$ and accumulates according to the law of motion $k_{t+1} = (1-\delta) k_t + i_t$. Capital used in the consumption sector is k_t^c and capital used in the investment sector is k_t^i. Aggregate labor input is $n_t = n_t^c + n_t^i$, where n_t^c is labor supplied in the consumption sector and n_t^i is labor supplied in the investment sector.

The representative household has period utility $u(c_t, 1-n_t)$ and preferences are time-separable with discount factor β. All markets are competitive and both labor and capital are free to move across sectors. Let p_t^c be the price of the consumption good and p_t^i be the price of the investment good.

1) Show that this economy aggregates into a one-sector growth model, i.e. the production structure can be summarized by an aggregate production technology $\tilde{z} f(k_t, n_t)$. What is the interpretation of the ratio $\left(p_t^c / p_t^i\right)$ in equilibrium?

Problem 1 Solution Our goal in this exercise is to summarize the two-sector production model

$$\begin{aligned} c &= z_c F(K_c, N_c) \\ i &= z_i F(K_i, N_i). \end{aligned}$$

by some aggregate production function of the form $y = \tilde{z} F(K, N)$ where $\tilde{z}$ does not depend on the distribution of capital and labor across the sectors.

In this question I will use two conventional notations:

- $K/N \equiv k$
- $F(k, 1) \equiv f(k)$

and denote aggregate values by:

- $K = K_c + K_i$, $N = N_c + N_i$.

CRS production implies the following:

- $F(K, N) = NF(K/N, N/N) = NF(K/N, 1) = Nf(k)$
- $F_K(K, N) = F_K(K/N, N/N)$

Now we are ready to look at the firm problem for each sector (which are quite similar).

I assume that households rent capital and labor to the firms. This assumption simplifies the solution because the firm need not take into account intertemporal considerations. The problem of the consumption sector firm is:

$$\underset{K_c,N_c}{Max}\left\{p_c z_c F\left(K_c,N_c\right)-K_c\left(r+\delta\right)-N_c w\right\}$$

where $(r+\delta)$ and w are the prices of capital and labor respectively.

The FOCs for the consumption sector are

$$\begin{aligned} p_c z_c F_k\left(K_c,N_c\right) &= r+\delta \\ p_c z_c F_n\left(K_c,N_c\right) &= w. \end{aligned}$$

Take the ratio of these and use the notation presented at the beginning of the question to write

$$\frac{f_k\left(k_c\right)}{f_n\left(k_c\right)}=\frac{r+\delta}{w},$$

Since the LHS is a strictly decreasing function of k, we can invert and write

$$k_c = g(\frac{r+\delta}{w})$$

Note that the RHS does not depend on sector specific values and therefore it follows that $k_i = g(\frac{r+\delta}{w})$ and $k_i = k_c$.

This equality together with $K = K_c + K_i$ and $N = N_c + N_i$ imply that: $k_i = k_c = k$ and therefore the capital labor ratio in each sector is equal to that of the (aggregate) economy.

The next step is to look at the aggregate production function. You should note that although f is the same for both sectors, different productivity shocks mean that we need to add two goods that in general can be very different in terms of their value (for example the consumption good can be seeds of wheat and the investment good can be machines). Therefore we have one degree of freedom in choosing how to add the two products. The choice I use here is to normalize the investment good to the consumption good by multiplying the level of investment good by $\frac{z_c}{z_i}$. We get:

$$\begin{aligned} y &= c + i\frac{z_c}{z_i} \\ &= z_c f(k) N_c + z_c f(k) N_i \\ &= z_c f(k)\left(N_c + N_i\right) \\ &= z_c F(K,N) \end{aligned}$$

The interpretation of the aggregate production is the total level of consumption goods and consumption-equivalent investment goods, which explains why the aggregate production function does not include z_i.

To get an expression for prices, take the ratio of the same FOC for two firms to get

$$\frac{p^c z^c}{p^i z^i} = 1 \Rightarrow \frac{p^i}{p^c} = \frac{z^c}{z^i},$$

so the relative prices measure investment-specific technical change.

For more on this, see Greenwood, Hercowitz and Krusell (1997) in the AER.

Remark. We have

$$p_c c + p_i i = \text{nominal GDP}.$$

Suppose we wanted to count GDP in real units of consumption; then we need to normalize by p_c. Let y denote exactly that: real GDP in units of consumption. Then

$$y = c + \frac{p^i}{p^c} i$$

If you substitute for consumption, investment good and prices the expressions that we derived above, you would get the exact same expression for y.